Model Development Report - Part 2

Training and Tuning | Objectives, validation and uncertainty

1. Training results: validation inside training only

EVS subset: 59,438 respondents / 36 countries; A170 has 59,122 valid responses. The fixed split yields 34,740 training, 5,007 production-validation, 10,079 same-country final-test and 9,296 held-out-country final-test cases. The 316 missing targets are excluded from predictive learning. Counts differ slightly from nominal percentages due to per-stratum rounding.

Held-out countries: AT, FR, GB, LT, MK, RS. Neither final prediction test was used for tuning or model comparison. Five within-country folds share country contexts but not respondents; five group folds remove entire training-country contexts.

Approach	Criterion	Fitted summary	Chosen tuning	Optimizer	CV MAE +/- SE Log loss RPS
baseline	Empirical absolute loss / frequencies	median 8; P(7-10)=0.726	None	Closed form	1.565 +/- 0.009 1.999 0.1251
ordinal	Threshold logistic + L2	81 coefficients + 9 cutpoints	alpha=0.01	L-BFGS-B; analytic gradient	1.255 +/- 0.007 1.802 0.1001
forest	Gini split gain	200 trees	depth 14; leaf 10; mtry 0.7	Greedy CART; seed 42	1.250 +/- 0.005 1.753 0.0987
boosting	Multiclass log loss	780 trees	lr 0.1; leaves 15; leaf 50; L2 1.0	Histogram boosting; early stop	1.225 +/- 0.006 1.715 0.0963
inference	Weighted ratio	p=0.712; SE=0.0030	No tuning; 1,000 bootstrap draws	Closed form; seed 42	N/A
causality	Not identified	Effect N/A; SE N/A	N/A: new design needed	Not fitted	N/A

Frozen selection: boosting; rule: lowest mean CV MAE. MAE uses the median of the predicted 10-category distribution. SE is the sample SD of five fold MAEs divided by sqrt(5): a descriptive tuning uncertainty, not a survey-design or independent-fold confidence interval.

2. Criteria and representation

Ordinal fitting minimizes all-threshold logistic surrogate loss plus L2; the baseline uses observed frequencies. Forests greedily improve Gini impurity; histogram boosting minimizes multinomial log loss. Final selection instead targets CV MAE; log loss and RPS check probability quality. No class balancing is applied, because it would change the learned prevalence. All predictive fitting is unweighted; survey-weighted inference answers a separate population question.

Age and age squared use fold-specific median imputation and scaling; a missing-age indicator is added. Nominal and ordinal survey items use explicit categories, including missingness. Country holdouts receive unknown-country indicator blocks during group CV; the public command rejects unsupported countries. The production artifact contains only 30 training-country contexts.

3. Alternative training criterion: squared versus Huber loss

Same numeric linear specification	CV MAE	CV MSE
OLS: half squared residual	1.2837	2.9393
Huber: delta fixed at 1 point	1.2731	2.9776

This audit supplement changes the global fitting loss, keeping the same unpenalized linear function class, training folds and support clipping (1-10). Squared loss targets a conditional mean; Huber limits tail influence and targets a robust centre. Numeric score distances are an explicit auxiliary assumption. Mean absolute OOF prediction change is 0.149 points (95th percentile 0.429). No test/production data or model reselection is involved.

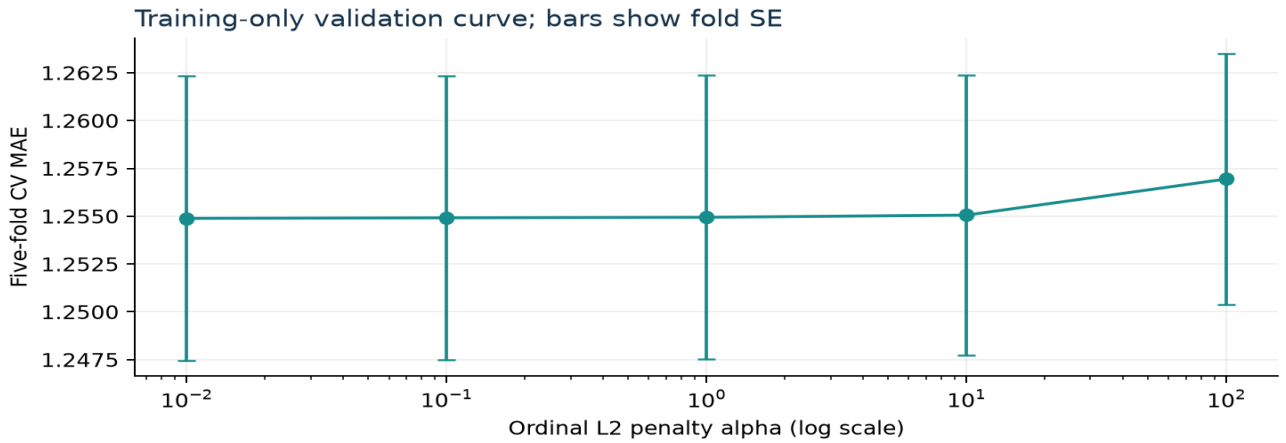
4. Search, chosen hyperparameters and sensitivity

Search methods: grid for ordinal alpha {0.1,1,10,100}; grid for forest depth {6,10} x minimum leaf {20,50} x max_features {sqrt,0.7}, with 200 trees; eight seeded random boosting configurations from learning_rate {0.05,0.1}, leaves {7,15}, minimum leaf {50,100}, L2 {0,1}. One boundary extension tried ordinal 0.01, forest depth 14/leaf 10 and boosting leaves 23/leaf 25. Selected settings are in the training table. Five identical folds refit all preprocessing; full trial values/timings are saved.

Boosting caps iterations at 200, uses 15% internal training-fold validation, patience 15, tol=1e-7 and seed 42. Additional audit diagnostics use three seeds and the same already-frozen configurations; they do not tune against production validation.

Frozen selected-family specification	CV MAE	Log loss	RPS
demographic_economic	1.447	1.892	0.1140
with_happiness	1.102	1.625	0.0870
demographics_only	1.470	1.909	0.1160

Sensitivity fits reuse the selected family's hyperparameters without retuning; differences assess predictor availability, not separately optimized competitors. Adding happiness tests near-proxy dependence; removing subjective inputs tests whether survey-like subjective answers are essential.



5. Optimization and fitted parameters

Ordinal: L-BFGS-B with analytic gradients, ordered-threshold increment constraints, initial slopes zero and cutpoints empirical cumulative logits; 2,000 initial iterations, ftol=1e-9, gtol=1e-5; retry at stricter tolerance when necessary. Final fit: 135 iterations; success=True; objective 100772.10; projected gradient per training row 2.8e-05. **Boosting:** final fit 78 iterations / 780 trees; internal validation log loss 1.9991 to 1.7282. Alternate-seed details are recorded in seed_stability.json.

Fitted ordinal parameters: largest absolute coefficients A009_5=-1.530; A173_10=+1.457; A009_1=+1.324; nine cutpoints [-3.77, -3.24, -2.55, -1.97, -0.98, -0.30, 0.68, 2.00, 3.07]. **Huber:** rank 73, start=OLS, L-BFGS-B, delta=1, maxiter=3000, ftol=1e-12, gtol=1e-7; 310 iterations, objective 0.8636, max gradient 2.5e-07; success=True. OLS is a closed-form rank-revealing least-squares solve. Both objectives are convex; multiple local starts are unnecessary.

Seeds 42/2026/2027: forest: CV MAE 1.2490-1.2514; full-training log loss best 1.5243, range 0.0009; boosting: CV MAE 1.2244-1.2286; full-training log loss best 1.6144, range 0.0053. These are stability checks of greedy randomized procedures, not a global optimum claim; no new model was selected.

Define continuity checks: the strictly demographic benchmark uses sex, age, education and country, excluding income/employment. All 30 training countries have OOF reliability bins in calibration_country_cv.csv (sparse bins flagged). The income gap is 34.24 pp at scores 7-10 and 34.75 pp at 8-10; the paid-work gap is 18.67 and 15.44 pp respectively. Weighted medians are 8 versus 7 for high/low income and 8 versus 7 for paid work/unemployment. The directions persist, not the gap sizes. Full 1-10 group distributions and threshold summaries are saved; these are descriptive training diagnostics, not new uncertainty or causal estimates.

Software/functions: custom OrdinalLogit.objective with scipy.optimize.minimize; sklearn.ensemble.RandomForestClassifier / HistGradientBoostingClassifier; scipy.linalg.lstsq and minimize for the criterion supplement. Complete defaults/non-defaults and fitted parameter arrays are saved as JSON. Greedy trees have no interchangeable continuous solver or global convergence certificate.

6. Transfer diagnostics and uncertainty

Frozen candidate	Grouped CV MAE	Log loss	RPS
baseline	1.565	2.003	0.1254
ordinal	1.282	1.822	0.1022
forest	1.281	1.782	0.1008
boosting	1.266	1.760	0.0997

Group CV is a secondary training-country diagnostic, not nested CV or a score on the six held-out test countries: configurations were chosen using within-country CV across the training sample. It exposes transfer to unfamiliar country indicator blocks. Country/subgroup errors and reliability bins are provided as aggregates. Predicted category probabilities express model uncertainty; no individual prediction intervals are claimed.

7. Population estimates and descriptive gaps

Overall weighting	High-satisfaction proportion	Approx. SE	Approx. 95% interval
unweighted	72.57%	0.22 pp	72.16-73.01%
equal_country	71.18%	0.30 pp	70.60-71.77%
population	67.14%	0.64 pp	65.94-68.42%

Descriptive contrast	Difference	Approx. SE	Approx. 95% interval
income high minus low	+34.24 pp	0.99 pp	+32.18 to +36.13 pp
paid minus unemployed	+18.67 pp	1.25 pp	+16.22 to +21.22 pp

Revised from Define: these estimates fit only the 34,740 training respondents in 30 countries. Analysis weights are gwght / pi, correcting the country x outcome partition retention (pi=0.5-0.7); pooled size factors define different summaries. Bootstrap intervals use 1,000 iid respondent resamples within fixed countries and fixed weights. They are approximate: national PSU dependence, target/income nonresponse and measurement bias remain. No inference to the six held-out countries is made. Groups with raw or effective n below 30 are suppressed.

Income is missing in 14.70% of the original EVS subset. For the revised training-country inference, size-weighted high-satisfaction proportions are 70.85% with observed income and 72.89% with missing income. This diagnoses a potentially selected complete-case population; it does not identify an MNAR correction.

8. Causality, reproducibility and remaining evaluation

Causal estimate: N/A; causal standard error: N/A. EVS has no baseline employment transition or 12-month outcome and cannot identify the intervention. The paid-work gap above is an association with a descriptive standard error, not a substitute causal estimate. A new longitudinal/experimental design is required; no estimator can repair absent identification.

Environment: Python 3.12.14; scikit-learn 1.9.1; NumPy 2.5.4; SciPy 1.18.1. Seed 42. Input Parquet SHA-256: 978592d03407591d74b1... Full source/code hashes, package versions, split policy and optimizer settings are recorded. Source: EVS/WVS ZA7505 v5.0.0, doi:10.4232/1.14320. Reproduce with make train, make inference and make develop-reports.